

Compilation of Summer Research Results

Brauch and Wiglesworth

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1 Stack Domination Density of Hypercubes

The following definition was introduced in [1].

Definition 1.1 (Stack Domination Density). *The quantity*

$$\bar{\gamma}(G) = \lim_{x \rightarrow \infty} \frac{\gamma(P_n \times G)}{n} = \lim_{x \rightarrow \infty} \frac{\gamma(C_n \times G)}{n}$$

*is the **stack domination density**.*

The following theorem was also introduced in [1]

Theorem 1.1. $\lim_{n \rightarrow \infty} \frac{\gamma(G \times P_n)}{n}$ and $\lim_{n \rightarrow \infty} \frac{\gamma(G \times C_n)}{n}$ either both converge or both diverge, and are equal if they converge.

Definition 1.2. *The product $C_k \times P_2$ is the prism graph $Y_{k,2}$. Each $Y_{k,2}$ has an interior and exterior cycle. Each vertex on the interior cycle will be labeled*

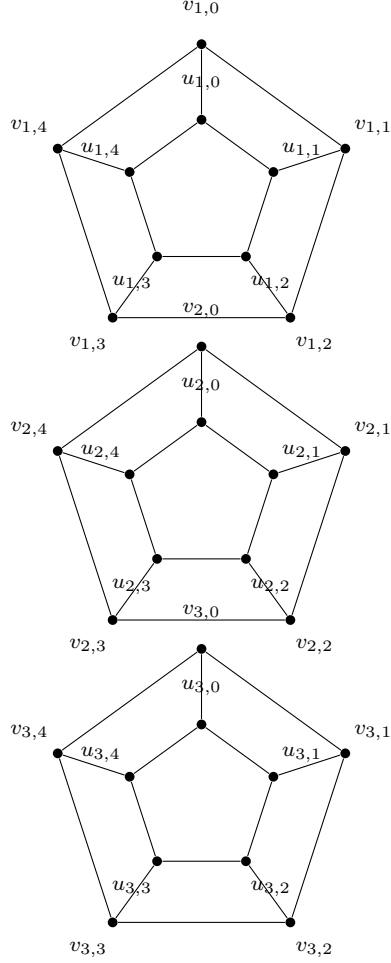
$$u_0, u_1, u_2, \dots, u_{k-1}$$

and vertices on the exterior cycle labeled

$$v_0, v_1, v_2, \dots, v_{k-1}$$

The edge set $E(Y_{k,2})$ contains $v_0v_1, \dots, v_{k-1}v_0$ and $u_0u_1, \dots, u_{k-1}u_0$ and $v_0u_0, \dots, v_{k-1}u_{k-1}$

Definition 1.3. *Let $G = Y_{k,2} \times P_n$ there are n fibers of $Y_{k,2}$. Using the same labeling for interior and exterior cycles as Definition 1.2 as well as a second subscript indicating which n^{th} fiber the vertices are apart of. For the r^{th} fiber the labeling for interior cycles would be $u_{r,0}, u_{r,1}, u_{r,2}, \dots, u_{r,k-1}$ and vertices on the exterior cycles labeled $v_{r,0}, v_{r,1}, v_{r,2}, \dots, v_{r,k-1}$ with $0 \leq r \leq n-1$. $E(G) = \{v_{r,i}v_{r,i+1} | 0 \leq i < k-1\} \cup \{u_{r,i}u_{r,i+1} | 0 \leq i < k-1\} \cup \{v_{r,i}u_{r,i} | 0 \leq i \leq k-1\} \cup \{v_{k-1}v_0, u_{k-1}u_0\}$.*



Lemma 1.1. *Let $G = Y_{k,2} \times P_n$, where $k > 3$ each vertex in an interior fiber has degree 5 because it is adjacent to 3 vertices within $Y_{k,2}$ and one vertex from the preceding and following fiber. Thus each vertex can dominate at most 6 vertices. Since $\bar{\gamma}$ is the ratio of dominating vertices per copy, the lowest bound is obtained when each vertex in the dominating set maximally dominates its neighboring vertices with no overlapping domination. Observe that a minimal lower bound can be obtained by dividing the total number of vertices, $2k$, per prism by 6,*

$$\bar{\gamma}(G) \geq \frac{2k}{6} \geq \frac{k}{3}$$

Thus the lowest possible stack domination density of G is $\frac{k}{3}$.

Theorem 1.2. $\bar{\gamma}(Q_1 \times P_n) = \frac{1}{2}$

Proof. PROOF HERE:

n	1	2	3	4	5	6	7	8	...
$\gamma(Q_1 \times P_n)$	1	2	2	3	3	4	4	5	$\lfloor n/2 \rfloor + 1$

Table 1: Observed domination numbers for the ladder graph $Q_1 \times P_n$ and the general pattern $\gamma(Q_1 \times P_n) = \lfloor n/2 \rfloor + 1$.

Without loss of generality, we split on the parity of n and verify that

$$\gamma(Q_1 \times P_n) = \begin{cases} \frac{n}{2} + 1, & \text{even } n, \\ \frac{n+1}{2}, & \text{odd } n. \end{cases}$$

We now evaluate

$$\lim_{n \rightarrow \infty} \frac{\gamma(Q_1 \times P_n)}{n}$$

- **Case 1 – n is even.** $\lim_{n \rightarrow \infty} \frac{\frac{n}{2} + 1}{n} = \lim_{n \rightarrow \infty} \frac{\frac{n+2}{2}}{n} = \lim_{n \rightarrow \infty} \frac{n+2}{2n} = \frac{1}{2}.$
- **Case 2 – n is odd.** $\lim_{n \rightarrow \infty} \frac{\frac{n+1}{2}}{n} = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2}.$

Hence, $\lim_{n \rightarrow \infty} \frac{\gamma(Q_1 \times P_n)}{n} = \frac{1}{2}.$

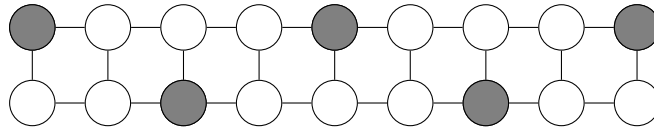


Figure 1: A ladder graph with 5 dominating vertices shaded in gray, having a domination density of $\frac{5}{9} \approx \frac{1}{2}$.

□

$$\lim_{x \rightarrow \infty} \frac{\gamma(G_{n,2})}{n} = \frac{1}{2}.$$

Pavlič and Žerovnik showed in [2] that $\gamma(P_n \times C_4) = 4$. We use this results below.

Theorem 1.3. $\bar{\gamma}(Q_2 \times P_n) = 1$

Proof. PROOF

n	1	2	3	4	5	6	7	...
$\gamma(Q_2 \times P_n)$	2	2	3	4	5	6	7	n

Table 2: Observed domination numbers for the ladder graph $Q_2 \times P_n$ and the general pattern $\gamma(Q_2 \times P_n) = n$ when n not equal to 1.

We now evaluate

$$\begin{aligned}
\bar{\gamma}(Q_2 \times P_n) &= \lim_{n \rightarrow \infty} \frac{\gamma(Q_2 \times P_n)}{n} \\
&= \lim_{n \rightarrow \infty} \frac{n}{n} \\
&= \lim_{n \rightarrow \infty} 1 \\
&= 1
\end{aligned}$$

Hence, $\lim_{n \rightarrow \infty} \frac{\gamma(Q_2 \times P_n)}{n} = 1$

□

Theorem 1.4. $\bar{\gamma}((C_4 \times P_2) \times P_n) = 1.5$

Proof. The *minimum* stack domination density *minimum* for $((C_4 \times P_2) \times P_n)$ can be proven by creating a sequence of 2 and 1 detailing the number of dominating vertices in each P_n .

$$...1, 2, 1, 2, 1...$$

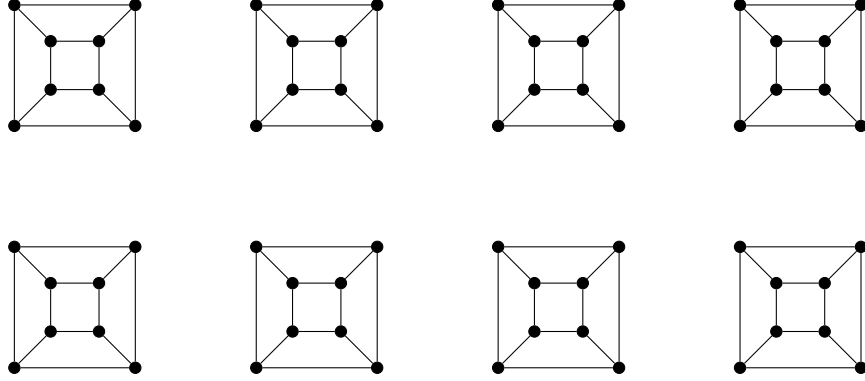
The dominating vertices can dominate 4 total vertices(including themselves), and dominate their corresponding vertices in their neighboring $C_4 \times P_2$, which is represented below:

$$\begin{array}{cccc}
...1, 2, & 1, & 2, 1... \\
1 & 4 & 1
\end{array}$$

Iterations of $C_4 \times P_2$ that have two dominating vertices can dominate all dominate themselves if the dominating vertices are picked well. $C_4 \times P_2$ with only one dominant vertex require the aid of two dominated vertices from both of their P_n neighbors to successfully dominate all vertices.

$$\begin{array}{c}
\dots 1, 2, 1, 2, 1\dots \\
\rightarrow 4 \leftarrow \\
+2 \\
+2
\end{array}$$

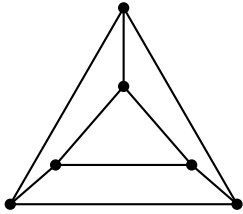
This creates a theoretical minimum for the *stack domination density*. Here is a cycle of $(C_4 \times P_2) \times P_n$ that proves the theoretical hypothesis.



□

Theorem 1.5. $\bar{\gamma}(C_3 \times P_2 \times P_n) = 1$

Proof. CHARLIE: argue why it can't be less than one and then show an example where it is one. In order for the infinite graph to have a stack domination density value of less than one, there would need to be a $C_3 \times P_2$ in the repeating pattern which does not contain a dominating vertex and yet still fully dominated. In other words it must be dominated from the connected prisms on either side. Each of those prisms would be have a single dominating vertex. Thus the total number of vertices being dominated within the prism without a dominating vertex could be 2, one coming each of the dominating vertices on the adjacent prisms. This leaves 4 undominated vertices in a prism, thus $\bar{\gamma}(C_3 \times P_2 \times P_n) \geq 1$



□

Theorem 1.6. $\bar{\gamma}(C_5 \times P_2 \times P_n) = 2$

Proof. JOSEPH: We first show that $\bar{\gamma}(C_5 \times P_2 \times P_n)$ cannot be less than 2.

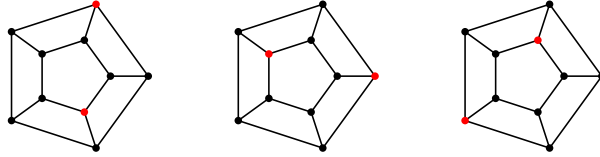
Each layer of the graph $C_5 \times P_2$ contains $5 \times 2 = 10$ vertices. Suppose a single layer has only 1 dominating vertex. That dominating vertex can cover itself and at most 3 adjacent vertices within its own layer, thus dominating up to 4 vertices in that layer. Furthermore, the two dominant vertices from each of the two adjacent layers are able to dominate 2 vertices each in the middle layer, contributing a total of 4 dominated vertices. However, the total domination from above and below would then cover at most 8 vertices, leaving 2 undominated in the layer.

Thus, a single dominating vertex is not enough for full domination. To ensure complete domination of each $C_5 \times P_2$ layer, we require at least 2 dominating vertices per layer.

Now, we construct a periodic domination sequence. The sequence of the number of dominating vertices per layer is:

$$\dots, 2, 2, 2, \dots$$

We choose two dominating vertices in each layer, where the minimum distance between them is 3 edges, and they are positioned in different P (P_1 and P_2). Through graphical verification, it can be shown that two well-placed dominating vertices are sufficient to fully dominate a $C_5 \times P_2 \times P_n$ layer. Moreover, the three layers together complete one full cycle.



□

Theorem 1.7. $\bar{\gamma}(C_6 \times P_2 \times P_n) = 2$

Proof. JOSEPH: We first show that $\bar{\gamma}(C_6 \times P_2 \times P_n)$ cannot be less than 2.

Each layer of the graph $C_6 \times P_2$ contains $6 \times 2 = 12$ vertices. Suppose a single layer has only 1 dominating vertex. That dominating vertex can dominate itself and at most 3 adjacent vertices within its own layer, thus covering up to 4 vertices locally. Additionally, the two dominating vertices from each of the two adjacent layers are able to dominate 2 vertices each in the middle layer, contributing a total of 4 more dominated vertices.

Therefore, at most $4 + 4 = 8$ vertices can be dominated in a layer that contains only one dominating vertex. Since the layer contains 12 vertices,

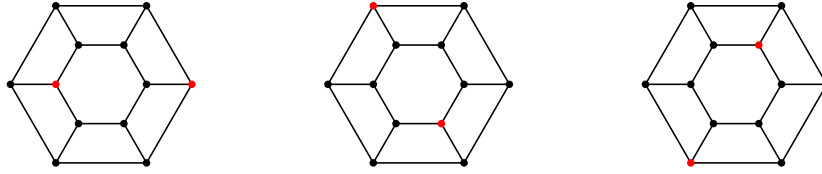
this leaves 4 undominated vertices. Thus, one dominating vertex is not sufficient for full domination.

To ensure complete domination of each $C_6 \times P_2$ layer, we require at least 2 dominating vertices per layer.

Now, we construct a periodic domination sequence. The sequence of the number of dominating vertices per layer is:

$$\dots, 2, 2, 2, \dots$$

We choose two dominating vertices in each layer, such that their minimum distance is 4 edges, and they are placed in different P layers (one in P_1 , one in P_2). Through graphical verification, it can be shown that two such dominating vertices can fully dominate an entire $C_6 \times P_2$ layer, and also help in dominating adjacent layers above and below. With this periodic pattern, three layers together form a complete cycle of domination.



□

Theorem 1.8. $\bar{\gamma}(C_8 \times P_2 \times P_n) = 3$

Proof. We first show that $\bar{\gamma}(C_8 \times P_2 \times P_n) = 3$ cannot be less than 3.

Each layer of the graph $C_8 \times P_2$ contains $8 \times 2 = 16$ vertices. Suppose that each fiber has only two dominating vertices. Those dominating vertices can dominate themselves and at most six adjacent vertices within its own fiber, thus covering up to eight vertices locally. Additionally, the three dominating vertices from each of the neighboring fibers are able to dominate six vertices in the middle fiber, contributing to a total of six more dominated vertices.

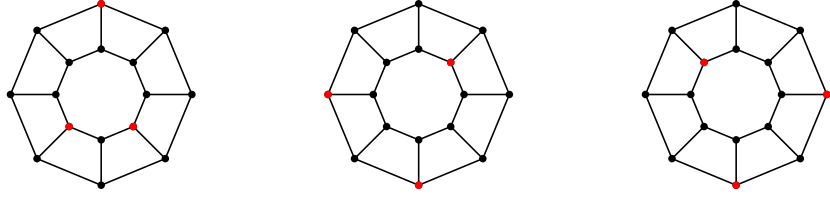
Therefore, at most $8 + 6 = 14$ vertices can be dominated in a layer that contains only two dominating vertices. Since the layer contains sixteen vertices, this leaves 2 undominated vertices. Thus, two dominating vertices are not sufficient for full domination in the middle fiber.

To ensure complete domination of each $C_8 \times P_2$ fiber, we require at least three dominant vertices per fiber.

Now, we construct a periodic domination sequence. The sequence of the number of dominating vertices per fiber is:

$$\dots, 3, 3, 3, \dots$$

We choose three dominating vertices in each fiber, such that the first and second, and first and third have a minimum distance of four edges and the second and third have to be on the same P layer. Through graphical verification, it can be shown that three such dominating vertices can fully dominate an entire $C_8 \times P_2$ fiber, and also help in dominating adjacent fibers above and below. With this periodic pattern, three fibers together form a complete cycle of domination.



□

Theorem 1.9. For $G = C_k \times P_2 \times P_n$, if $k \equiv 0 \pmod{6}$, then $\bar{\gamma}(G) = k/3$

Proof. Let $G = C_k \times P_2 \times P_n$, $k \equiv 0 \pmod{6}$ From Lemma 1.1 the lowest possible stack domination density of G is $k/3$. Define $D \subseteq V(G)$ according to the following. Using the labeling given in definition 1.3. Add vertices to the set D according to the following. For each r^{th} fiber of $Y_{k,2}$ in G :

- If $r \equiv 0 \pmod{3}$, add $v_{r,s}$ where $s \equiv 3 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 0 \pmod{6}$.
- If $r \equiv 1 \pmod{3}$, add $v_{r,s}$ where $s \equiv 1 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 4 \pmod{6}$.
- If $r \equiv 2 \pmod{3}$, add $v_{r,s}$ where $s \equiv 5 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 2 \pmod{6}$.

Thus for every copy, $\frac{k}{3}$ vertices will be added. The neighborhood $N(d)$ for all $d \in D$ are of magnitude 5 since G is regular. All of these neighborhoods will be disjoint if the path between each $d \in D$ is at minimum length 3. In order for this to be true there cannot exist a path of length 2 between dominating vertices. This is true within a fiber since the vertices chosen will always be alternating between interior and exterior cycles with difference in labeling of 3. A path between dominating vertices in adjacent fibers cannot be of length 2 since you must first traverse to the adjacent fiber, then for any dominated vertex u or v the nearest other dominating vertex will be 2 vertices away. Thus for all of the nearest vertices a path will be at minimum

of length 3. All of the neighborhoods are disjoint and of cardinality 5. Thus each vertex in the dominating set maximally dominates with no vertex being dominated twice. Observe that the total number of dominated vertices will be

$$|D| * 6 = \frac{k}{3} * 6 = 2k$$

, which is the total number of vertices in each copy. Every vertex in G will be dominated by D , thus $\bar{\gamma}(G) = k/3$ $\square$

Conjecture 1.1. *For $G = C_k \times P_2 \times P_n$, if $k \equiv 3 \pmod{6}$, then $\bar{\gamma}(G) = k/3$*

Proof. For this graph $G = C_k \times P_2 \times P_n$, $k \equiv 0 \pmod{6}$, each vertex is adjacent to 3 vertices within $C_k \times P_2$, and 2 vertices from the copy's on either side. Thus each vertex can dominate at most 6 vertices including itself. Since $\bar{\gamma}$ is the ratio of dominating vertices per copy, the lowest possible value is obtained when each vertex maximally dominates its surrounding vertices, with no overlapping domination. Thus, a minimal lower bound can be obtained by dividing the total number of vertices, $2k$, per prism by 6,

$$\bar{\gamma}(G) \geq 2k/6 \geq k/3$$

Thus the lowest possible stack domination density of G is $k/3$. We will now provide a construction for all G which satisfies this lowest bound.

Each $C_k \times P_2$ is a prism $Y_{k,2}$ with an interior and exterior cycle. There are n copies of these $Y_{k,2}$.

Each vertex on the interior cycles of $Y_{k,2}$ will be labeled

$$u_{r,0}, u_{r,1}, u_{r,2}, \dots, u_{r,k-1}$$

and vertices on the exterior cycles labeled

$$v_{r,0}, v_{r,1}, v_{r,2}, \dots, v_{r,k-1}$$

where r denotes which copy of $Y_{k,2}$, with $0 \leq r \leq n-1$.

The edge set contains $v_{r,i}, v_{r,i+1}$ and $u_{r,i}, u_{r,i+1}$ for each $i < k-1$, as well as $v_{r,k-1}, v_{r,0}$ and $u_{r,k-1}, u_{r,0}$, and $u_{r,j}, v_{r+1,j}, u_{r,j}, v_{r+1,j}$ belongs in $E(G)$.

Add vertices to the set D according to the following. For each r the fibre of $Y_{k,2}$ in G :

- If $r \equiv 0 \pmod{8}$, add $v_{r,s}$ where $s \equiv 0 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 0 \pmod{6}$.

- If $r \equiv 1 \pmod{8}$, add $v_{r,s}$ where $s \equiv 1 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 4 \pmod{6}$.
- If $r \equiv 2 \pmod{8}$, add $v_{r,s}$ where $s \equiv 5 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 2 \pmod{6}$.
- If $r \equiv 3 \pmod{8}$, add $v_{r,s}$ where $s \equiv 3 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 0 \pmod{6}$.
- If $r \equiv 4 \pmod{8}$, add $v_{r,s}$ where $s \equiv 1 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 4 \pmod{6}$.
- If $r \equiv 5 \pmod{8}$, add $v_{r,s}$ where $s \equiv 5 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 2 \pmod{6}$.
- If $r \equiv 6 \pmod{8}$, add $v_{r,s}$ where $s \equiv 1 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 4 \pmod{6}$.
- If $r \equiv 7 \pmod{8}$, add $v_{r,s}$ where $s \equiv 5 \pmod{6}$ as well as $u_{r,t}$ where $t \equiv 2 \pmod{6}$.

Thus for every copy, $\frac{k}{3}$ vertices will be added.

The neighborhood $N(d)$ for all $d \in D$ are disjoint and are all of magnitude 5. All of these neighborhoods will be disjoint if and only if the shortest path between all vertices is atleast 3. All three possible iterations of the constructions ($r = 0 \bmod 8, 1 \bmod 8, 2 \bmod 8, 3 \bmod 8, 4 \bmod 8, 5 \bmod 8, 6 \bmod 8, 7 \bmod 8$) no dominating vertices is within two vertices of each other. For any $u_{r,i}, v_{r,i}$ the vertices $u_{r+1,i}, v_{r+1,i}$ are not adjacent to another vertex in D so the path between these vertices must be atleast 3. Thus all of the neighborhoods are disjoint and of cardinality 5. Thus each vertex in the dominating set maximally dominates with no vertex being dominated twice, thus the stack domination density is equivalent to the lowest bound found earlier of $k/3$. $\square$

Algorithm 1.1. Clockwise Algorithm for $k \equiv 0 \pmod{3}$. For fiber r , if r is odd, begin by adding vertex $v_{r,r-1 \pmod{m}}$ to your set. If r is even, begin by adding vertex $u_{r,r-1 \pmod{m}}$ to your set.

While the number of vertices for a potential dominating set in fiber r is less than $\frac{m}{3}$:

- if the most recent vertex added is $v_{r,i}$ the next vertex to add to the set is $u_{r,(i+3) \pmod{m}}$.

- if the most recent vertex added is $u_{r,i}$ the next vertex to add to the set is $v_{r,(i+3) \bmod m}$.

When you have $\frac{m}{3}$ vertices the fiber is completed.

Theorem 1.10. *Let $G = C_m \times P_2 \times P_n$. If $m \equiv 0 \pmod{3}$, then the stack domination density satisfies*

$$\bar{\gamma}(G) = \frac{m}{3}.$$

Proof. Let $G = C_m \times P_2 \times P_n$, where $m \equiv 0 \pmod{3}$. From Lemma 1.1, the lowest possible stack domination density of G is $\frac{m}{3}$.

Define $D \subseteq V(G)$ according to Algorithm 1.1.

To confirm that D is a minimal dominating set, we require that every vertex in G is dominated exactly once. That is, $|D| \cdot 6 = |V(G)|$, and the closed neighborhoods of vertices in D must be pairwise disjoint.

Because G is 5-regular, each vertex in D dominates 6 vertices, its 5 neighbors and itself. Thus, the total number of dominated vertices is at most

$$|D| \cdot 6 = \left(\frac{mn}{3}\right) \cdot 6 = 2mn = |G|,$$

. To show that D dominates exactly this many vertices, we will show that no vertex has two neighbors in D , meaning that each vertex is dominated by exactly one vertex in D . To do this, we will argue that $d(x, y)$, where $x, y \in D$ is at most 3.

By the construction of D given in Algorithm 1.1, observe that if $u_{r,i} \in D$, then neither $u_{r+1,i}$ nor $u_{r+2,i}$ is in D . Similarly, if $v_{r,j} \in D$, then neither $v_{r+1,j}$ nor $v_{r+2,j}$ is in D . Therefore, we only need to consider cases where x, y are either in the same fiber or in adjacent fibers.

- Case 1: Let x, y belong to the same fiber. Since the algorithm alternates between adding vertices on the interior and exterior cycles of the same fiber of $Y_{m,2}$ and advances by 3 positions with each step, any two vertices in D within a fiber differ by both cycle and position, ensuring a path length of at least 3 between them.
- Case 2: Let x, y be in adjacent fibers. From the construction, each vertex in fiber $r + 1$ changes all three components (cycle, fiber, and position) relative to those in fiber r . Thus, the shortest path between any such x and y is again at least 3.

Thus the intersection of the sets of vertices dominated by any two vertices in D is empty. Therefore, D dominates exactly $2mn = |G|$ vertices. Therefore,

$$\lim_{n \rightarrow \infty} \frac{\gamma(Y_{m,2} \times P_n)}{n} = \frac{|D|}{n} = \frac{mn}{3n} = \frac{m}{3}$$

□

Theorem 1.11. *If $m \equiv 0 \pmod{3}$, $\bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m}{3}$*

Proof. Assume one fiber has less than $\frac{m}{3}$, $\frac{m}{3} - 1 = \frac{m-3}{3}$ dominating vertices. Vertices inside the fiber can only dominate four vertices in itself. The fiber also receives help from both neighbors with at least two vertices. The expression can be formed as: $4(\frac{m-3}{3}) + 2(\frac{m}{3})$. Performing algebra, we simplify.

$$\frac{4m-12}{3} + \frac{2m}{3} = \frac{6m-12}{3} = 2m - 4$$

The result is inadequate because there needs to be a total of $2m$ dominated vertices in $C_k \times P_2$. This leaves us with two options: (1) four dominating vertices must be selected from the neighbors, or (2) an additional vertex within the current fiber must provide domination. The expression below details the second option's algebra process.

$$\frac{\frac{m-3}{3} + \frac{m}{3} + 1}{2} = \frac{\frac{m-3+m+3}{3}}{2} = \frac{\frac{2m}{3}}{2} = \frac{2m}{3} \times \frac{1}{2} = \frac{m}{3}$$

With only adding one vertex to the dominating set it equals $\frac{m}{3}$, but the expression will be greater than $\frac{m}{3}$ if more vertices from the fiber's neighbors are added changing the +1 to +2 or more.

$$\therefore \text{ if } m \equiv 0 \pmod{3}, \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m}{3}$$

□

Theorem 1.12. *If $m \equiv 1 \pmod{3}$, $\bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{2m+1}{6}$*

Proof. Assume one fiber has less than $\frac{2m+1}{6}$, $\frac{2m+1}{6} - 1 = \frac{2m-5}{6}$ dominating vertices. Vertices inside the fiber can only dominate four vertices in itself. The fiber also receives help from both neighbors with at least two vertices. The expression can be formed as: $4(\frac{2m-5}{6}) + 2(\frac{2m+1}{6})$. Performing algebra, we simplify.

$$\frac{8m-20}{6} + \frac{4m+2}{6} = \frac{12m-18}{6} = 2m - 3$$

The result is inadequate because there needs to be a total of $2m$ dominated vertices in $C_k \times P_2$. This leaves us with two options: (1) three dominating vertices must be selected from the neighbors, or (2) an additional vertex within the current fiber must provide domination. The expression below details the second option's algebra process.

$$\frac{\frac{2m-5}{6} + \frac{2m+1}{6} + 1}{2} = \frac{\frac{2m-5+2m+1+6}{6}}{2} = \frac{\frac{4m+2}{6}}{2} = \frac{4m+2}{6} \times \frac{1}{2} = \frac{2m+1}{6}$$

With only adding one vertex to the dominating set it equals $\frac{2m+1}{6}$, but the expression will be greater than $\frac{2m+1}{6}$ if more vertices from the fiber's neighbors are added changing the +1 to +2 or more.

$$\therefore \text{ if } m \equiv 1 \pmod{3}, \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{2m+1}{6}$$

□

From Jonah. The above theorem requires more work since it basically means that a dominating set can have a decimal number of vertices. □

Theorem 1.13. *If $m \equiv 2 \pmod{3}$, $\bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m+1}{3}$*

Proof. Assume one fiber has less than $\frac{m+1}{3}$, $\frac{m+1}{3} - 1 = \frac{m-2}{3}$ dominating vertices. Vertices inside the fiber can only dominate four vertices in itself. The fiber also receives help from both neighbors with at least two vertices. The expression can be formed as: $4(\frac{m-2}{3}) + 2(\frac{m+1}{3})$. Performing algebra, we simplify.

$$\frac{4m-8}{3} + \frac{2m+2}{3} = \frac{6m-6}{3} = 2m - 2$$

The result is inadequate because there needs to be a total of $2m$ dominated vertices in $C_k \times P_2$. This leaves us with two options: (1) two dominating vertices must be selected from the neighbors, or (2) an additional vertex within the current fiber must provide domination. The expression below details the second option's algebra process.

$$\frac{\frac{m-2}{3} + \frac{m+1}{3} + 1}{2} = \frac{\frac{m-2+m+1+3}{3}}{2} = \frac{\frac{2m+2}{3}}{2} = \frac{2m+2}{3} \times \frac{1}{2} = \frac{m+1}{3}$$

With only adding one vertex to the dominating set it equals $\frac{m+1}{3}$, but the expression will be greater than $\frac{m+1}{3}$ if more vertices from the fiber's neighbors are added changing the +1 to +2 or more.

$$\therefore \text{ if } m \equiv 2 \pmod{3}, \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m+1}{3}$$

□

above 3 theorems edited below

Theorem 1.14. *For $G = C_m \times P_2 \times P_n$, if $m \equiv 0 \pmod{3}$, then $\bar{\gamma}(G) \geq \frac{m}{3}$*

Proof. Assume one fiber has less than $\frac{m}{3}$ dominating vertices, or $\frac{m}{3} - 1 = \frac{m-3}{3}$ dominating vertices. Vertices inside the fiber can only dominate four vertices within $Y_{m,2}$. The fiber is also dominated from both neighboring fibers with at least two vertices. The expression representing total dominated vertices within a fiber can be written as: $4(\frac{m-3}{3}) + 2(\frac{m}{3})$. Performing algebra, we simplify.

$$\frac{4m-12}{3} + \frac{2m}{3} = \frac{6m-12}{3} = 2m - 4$$

G has a total of $2m$ vertices, thus this fiber will need 4 more vertices to be dominated, leaving us with two possibilities.

Case 1: If more dominating vertices are added to the neighboring fibers, the total will be greater than $\frac{m}{3}$ since 4 more vertices would have to be added, increasing the stack domination density to offset the fiber with less dominating vertices.

Case 2: If an additional vertex is added within the fiber than this fiber does not have less than $\frac{m}{3}$ vertices, thus the total stack domination density must be greater than or equal to $\frac{m}{3}$

$$\therefore \text{ if } m \equiv 0 \pmod{3}, \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m}{3}$$

□

Theorem 1.15. *For $G = C_m \times P_2 \times P_n$, if $m \equiv 1 \pmod{3}$, then $\bar{\gamma}(G) \geq \frac{2m+1}{6}$*

Proof. $\bar{\gamma}(G)$ cannot be less than $\frac{2m+1}{6}$ unless there exists a fiber with less than $\frac{2m+1}{6}$ dominating vertices. Suppose, for contradiction, that a fiber in $C_m \times P_2 \times P_n$ contains fewer than $\frac{2m+1}{6}$ dominating vertices. Then it contains at most $\frac{2m+1}{6} - 1 = \frac{2m-5}{6}$ dominating vertices.

Each dominating vertex can dominate at most 4 vertices within the same fiber. Thus, these $\frac{2m-5}{6}$ dominating vertices dominate at most $4 \cdot \frac{2m-5}{6}$ vertices in the fiber. Additionally, each neighboring fiber must contribute at least two vertices to help dominate the current fiber, accounting for an extra $2 \cdot \frac{2m+1}{6}$ dominated vertices. Altogether, this gives:

$$4 \cdot \frac{2m-5}{6} + 2 \cdot \frac{2m+1}{6} = \frac{8m-20}{6} + \frac{4m+2}{6} = \frac{12m-18}{6} = 2m - 3.$$

However, each fiber contains exactly $2m$ vertices, all of which must be dominated. Since $2m - 3 < 2m$, this count is insufficient. Thus, our assumption must be false.

To ensure domination of all $2m$ vertices in the fiber, either: (1) An additional dominating vertex must be placed within the fiber, contradicting our assumption, or (2) More vertices must be selected from the neighboring fibers.

Suppose we add one more dominating vertex within the fiber. Then the new average number of dominating vertices is:

$$\frac{\left(\frac{2m-5}{6} + \frac{2m+1}{6} + 1\right)}{2} = \frac{\frac{4m+2}{6}}{2} = \frac{2m+1}{6}.$$

Adding more than one additional vertex would increase this value beyond $\frac{2m+1}{6}$. Therefore, to dominate all vertices, at least $\frac{2m+1}{6}$ dominating vertices are required on average per fiber.

$$\therefore \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{2m+1}{6} \quad \text{when } m \equiv 1 \pmod{3}.$$

□

From Jonah. The above theorem requires more work since it basically means that a dominating set can have a decimal number of vertices. □

Theorem 1.16. *For $G = C_m \times P_2 \times P_n$, if $m \equiv 2 \pmod{3}$, then $\bar{\gamma}(G) \geq \frac{m+1}{3}$*

Proof. Suppose, for contradiction, that a fiber in $C_m \times P_2 \times P_n$ contains fewer than $\frac{m+1}{3}$ dominating vertices. Then it contains at most

$$\frac{m+1}{3} - 1 = \frac{m-2}{3} \text{ dominating vertices.}$$

Each dominating vertex can dominate at most 4 vertices within the same fiber. Thus, these $\frac{m-2}{3}$ dominating vertices cover at most

$$4 \cdot \frac{m-2}{3}$$

vertices in the fiber. Additionally, each neighboring fiber must contribute at least two vertices to help dominate the current fiber, accounting for an extra $2 \cdot \frac{m+1}{3}$ dominated vertices. Altogether, this gives:

$$4 \cdot \frac{m-2}{3} + 2 \cdot \frac{m+1}{3} = \frac{4m-8}{3} + \frac{2m+2}{3} = \frac{6m-6}{3} = 2m-2.$$

However, each $C_m \times P_2$ fiber contains exactly $2m$ vertices, all of which must be dominated. Thus, our assumption must be false.

To ensure domination of all $2m$ vertices in the fiber, either: (1) An additional dominating vertex must be placed within the fiber, or (2) More vertices must be selected from the neighboring fibers.

Suppose we add one more dominating vertex within the fiber. Then the new average number of dominating vertices is:

$$\frac{\frac{m-2}{3} + \frac{m+1}{3} + 1}{2} = \frac{\frac{m-2+m+1+3}{3}}{2} = \frac{\frac{2m+2}{3}}{2} = \frac{2m+2}{3} \times \frac{1}{2} = \frac{m+1}{3}$$

Adding more than one additional vertex would increase this value beyond $\frac{m+1}{3}$. Therefore, to dominate all vertices, at least $\frac{m+1}{3}$ dominating vertices are required on average per fiber.

$$\therefore \bar{\gamma}(C_m \times P_2 \times P_n) \geq \frac{m+1}{3} \quad \text{when } m \equiv 2 \pmod{3}.$$

□

end of edited thms

Algorithm 1.2. Clockwise Algorithm for $k \equiv 2 \pmod{3}$ For fiber r , if r is odd, begin by adding vertex $v_{r,r-1 \pmod{m}}$ to your set. If r is even, begin by adding vertex $u_{r,r-1 \pmod{m}}$ to your set. While the number of vertices for a potential dominating set in fiber r is less than $\frac{m+1}{3}$:

- if the most recent vertex is $v_{r,i}$ the next vertex to add to the set is $u_{r,(i+3) \pmod{m}}$.
- if the most recent vertex is $u_{r,i}$ the next vertex to add to the set is $v_{r,(i+3) \pmod{m}}$.

When you have $\frac{m+1}{3}$ vertices the fiber is completed.

For example, in $Y_{8,2} \times C_3$, a potential clockwise dominating set is for the first fiber is $v_{1,0} \rightarrow u_{1,3} \rightarrow v_{1,6}$.

Theorem 1.17. For the graph $Y_{m,2} \times P_n$ where $m \equiv 2 \pmod{3}$, a dominating set can be constructed using the Algorithm 1.2. This dominating set will require $\frac{m+1}{3}$ vertices per fiber.

Proof. Let D be the set of vertices defined by Algorithm 1.2. Let $v_{r,s}$ be an arbitrary vertex on the exterior cycle. If $v_{r,s} \in D$, then $v_{r,s}$ is dominated. Suppose not, $v_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$, then $s \not\equiv 4 \pmod{6}$

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r+1),s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r+2) \pmod{3},s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,(s+1)}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$

Case 2: $r \equiv 1 \pmod{3}$, then $s \not\equiv 0 \pmod{6}$

- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,(s-1)}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,(s+1)}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r+1),s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r-1),s}$

Case 3: $r \equiv 2 \pmod{3}$, then $s \not\equiv 2 \pmod{6}$

- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,(s+1)}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,(s-1)}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r+1),s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{(r-1),s}$

For all s , $v_{r,s}$ is dominated.

Let $u_{r,s}$ be an arbitrary vertex on inner cycle. If $u_{r,t} \in D$, then $u_{r,s}$ is dominated. Suppose not, $u_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$, then $t \not\equiv 1 \pmod{6}$

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,(t-1)}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{(r+1) \pmod{3},t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+2 \pmod{3},t}$

Case 2: $r \equiv 1 \pmod{3}$, then $t \not\equiv 3 \pmod{6}$

- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,(t+1)}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,(t+1)}$
- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$

- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{(r+1),t}$

Case 3: $r \equiv 2 \pmod{3}$, then $t \not\equiv 5 \pmod{6}$

- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,(t+1)}$
- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,(t-1)}$ (with the exception of $u_{r,0}$ which is always dominated by $v_{r,0}$)
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{(r+1),t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $v_{(r-1),t}$

For all t , $u_{r,t}$ is dominated. Since for every possible

□

$m \bmod 6$	n
0	3
1	$4m$
2	3
3	$2m$
4	$LCM(2m, 8(1 + \lfloor \frac{m}{6} \rfloor))$
5	3

Table 3: Number of copies of prisms before dominating sets repeat.

2 Unified Results

Definition 2.1. *fiber_k*: In a stacked prism, $C_m \times P_2 \times P_n$ or $C_m \times P_2 \times C_n$, let *fiber_k* be the set of vertices in the k^{th} fiber that should be used to construct a dominating set in the stack prism.

Conjecture 2.1. For any value m of $(C_m \times P_2 \times P_n)$, the n amount of fibers required to repeat creating a cycle can be calculated based on the value m .

[From Jonah] Joseph is on the right track based on the photo he sent with the table of prediction $m \equiv 1 \bmod 3$, although I don't know where to take it from there to prove this table.

Conjecture 2.2. For any $m \equiv 1, 3, 4 \bmod 6$

Lemma 2.1. For $m \equiv 1 \bmod 6$ in $(C_m \times P_2 \times P_n)$, $n = 4m$.

Proof. In each copy of C_m , there exist $2m$ vertices and coupled with the clockwise algorithm there exist $2m$ configurations using each vertex as a starting point with $|fiber_k| = \frac{m+2}{3}$. For the exception of $m \equiv 1 \bmod 6$, there exist $2m$ vertices coupled with the clockwise algorithm makes $2m$ configurations using each vertex as a starting point with $|fiber_k| = \frac{m-1}{3}$. In order to form a cycle on $(C_m \times P_2 \times P_n)$, the configurations must form an alternating pattern of $|fiber_k| = \frac{m+2}{3}$, $|fiber_{k+1}| = \frac{m-1}{3}$, $|fiber_{k+2}| = \frac{m+2}{3}$, etc. Every configuration must be used at least once from both sets of configurations so it will take a minimum of $n = 2m + 2m = 4m$ fibers. $\square$

Lemma 2.2. For $m \equiv 3 \bmod 6$ in $(C_m \times P_2 \times P_n)$, $n = 2m$.

Proof. In each copy of C_m , there exist $2m$ vertices and coupled with the clockwise algorithm there exist $2m$ configurations using each vertex as a starting point. In order to form a cycle on $(C_m \times P_2 \times P_n)$, every configuration must be used at least once, so there exist an order of fibers such that $n = 2m$. $\square$

Lemma 2.3. For $m \equiv 4 \pmod 6$ in $(C_m \times P_2 \times P_n)$, $n = LCM(2m, 8(1 + \lfloor \frac{m}{6} \rfloor))$.

Proof. When finding a dominating set for $m \equiv 4 \pmod 6$ and following the pattern of $fiber_k, fiber_{k+1}, fiber_{k+2} \dots$ and $\frac{m+2}{3}, \frac{m-1}{3}, \frac{m+2}{3}, \dots$, there are only $\lceil \frac{m}{3} \rceil$ viable patterns for $fiber_k$ containing $\frac{m+2}{3}$ vertices that are optimal.

There are $2m$ vertices in fiber k . $\square$

Algorithm 1 Clockwise Algorithm

Require: $m \geq 3$, $startPoint$

Ensure: $|Set| = \lceil \frac{m}{3} \rceil$

$Set \leftarrow [startPoint]$

$position \leftarrow startPoint$

while $|Set| < \lceil \frac{m}{3} \rceil$ **do**

if $position$ is $v_{r,i}$ **then**

 append $u_{r,(i+3) \bmod m}$ to Set

$position \leftarrow u_{r,(i+3) \bmod m}$

else if $position$ is $u_{r,i}$ **then**

 append $v_{r,(i+3) \bmod m}$ to Set

$position \leftarrow v_{r,(i+3) \bmod m}$

end if

end while

In Figure 2 we apply the clockwise algorithm (Algorithm 1) to the prism graph $(C_8 \times P_2)$ starting with $v_{r,0}$. The next vertex added to the set is $u_{r,(0+3) \bmod 8} = u_{r,3}$. Then the vertex $v_{r,(3+3) \bmod 8} = v_{r,6}$ is added to the set. The set now has $\lceil \frac{8}{3} \rceil = 3$ vertices, so then the algorithm ends. These vertices are highlighted in Figure 2.

In Figure 3 we apply the domination algorithm (Algorithm 2) to the prism graph $(C_3 \times P_2)$. Starting with fiber one, we apply the clockwise algorithm (Algorithm 1) starting with $v_{1,0}$. Since $m = 3$, the clockwise algorithm will always produce a set with $\lceil \frac{3}{3} \rceil = 1$ vertex in each fiber. The next step in the domination algorithm is to construct fiber two. Because

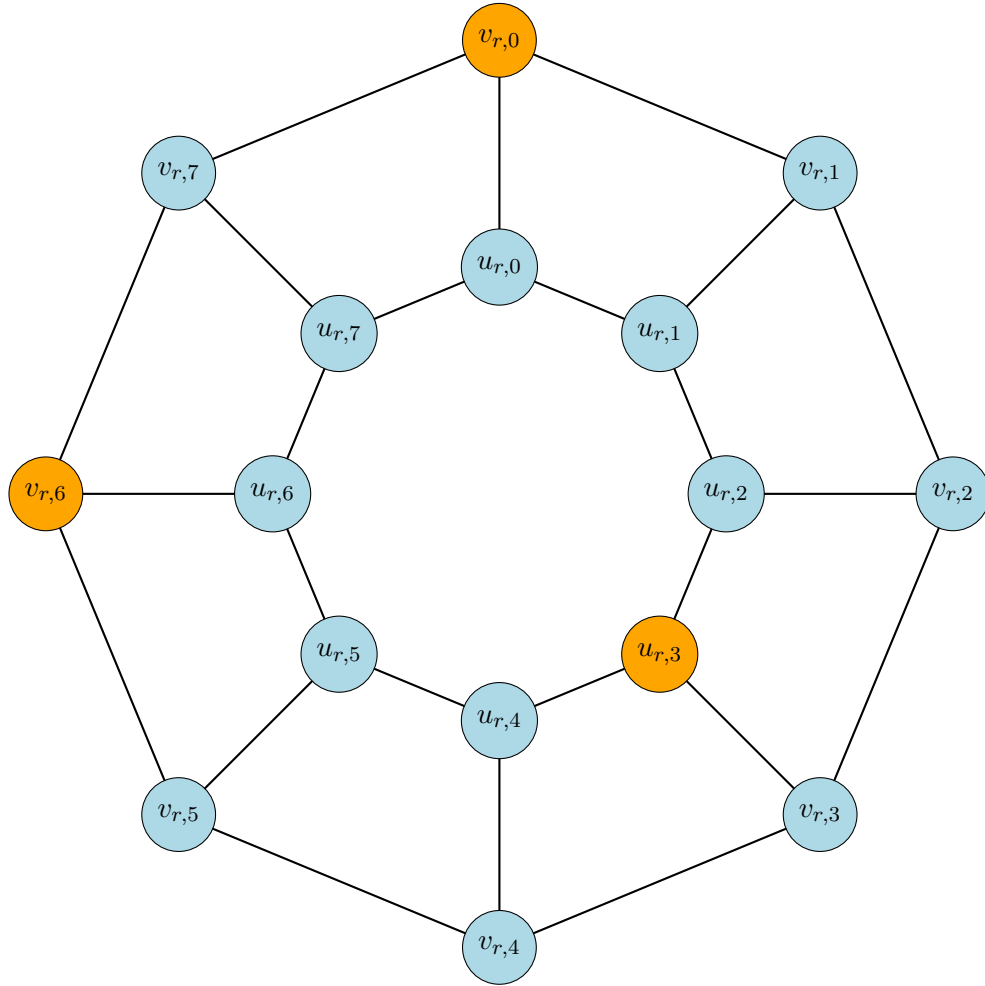


Figure 2: Application of Clockwise Algorithm (Algorithm 1) on $m = 8$ and starting at $v_{r,0}$

Algorithm 2 Domination Algorithm 0, 1, 3, 4 mod 6

Require: $m \geq 3$

Ensure: *dominatingSet*

$fiber_1 \leftarrow \text{clockwise}(m, \text{startPoint} = v_{1,0})$

append $fiber_1$ to *dominatingSet*

if $m \equiv 0, 1, 4 \pmod{6}$ **then**

$fiber_2 \leftarrow \text{clockwise}(m, \text{startPoint} = v_{2,2})$

else if $m \equiv 3 \pmod{6}$ **then**

$fiber_2 \leftarrow \text{clockwise}(m, \text{startPoint} = u_{2,m-1})$

end if

if $m \equiv 1 \pmod{3}$ **then**

Remove last vertex of $fiber_2$

end if

append $fiber_2$ to *dominatingSet*

$currentFiber \leftarrow fiber_2$

while $currentFiber \neq fiber_1$ **do**

$nextFiber \leftarrow$ undominated vertices from *previousFiber*

if $nextFiber \neq fiber_1$ **then**

append $nextFiber$ to *dominatingSet*

end if

$currentFiber \leftarrow nextFiber$

end while

Algorithm 3 Domination Algorithm 0, 2, 5 mod 6

Require: $m \geq 3$

Ensure: *dominatingSet*

$fiber_1 \leftarrow \text{clockwise}(m, \text{startPoint} = v_{1,0})$

append $fiber_1$ to *dominatingSet*

$fiber_2 \leftarrow \text{clockwise}(m, \text{startPoint} = v_{2,2})$

append $fiber_2$ to *dominatingSet*

$fiber_3 \leftarrow \text{clockwise}(m, \text{startPoint} = u_{3,1})$

append $fiber_3$ to *dominatingSet*

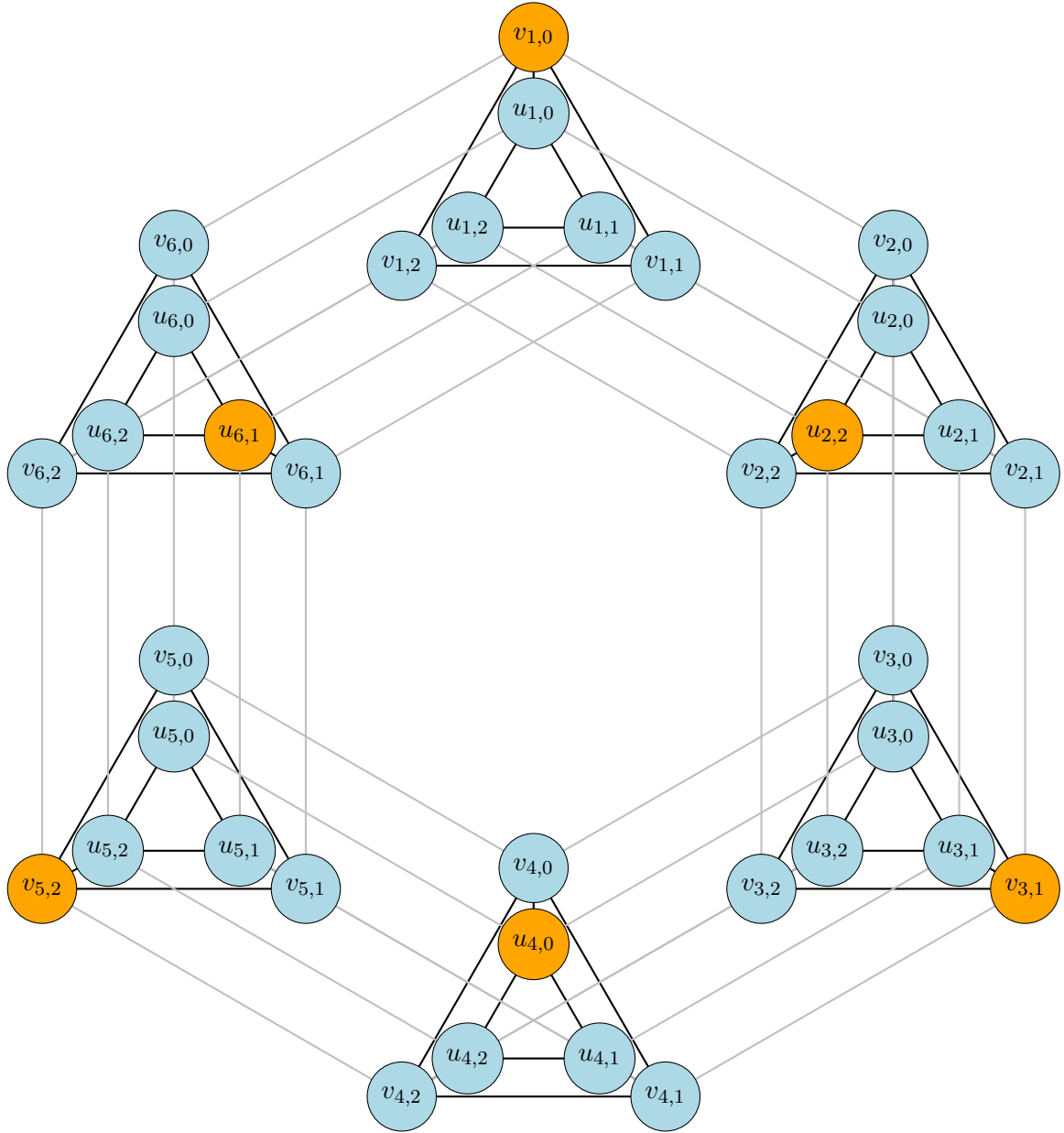


Figure 3: Application of Domination Algorithm (Algorithm 2) on $m = 3$

$m \equiv 3 \pmod{6}$, the clockwise algorithm starts at $u_{2,m-1} = u_{2,2}$. Now we enter the while loop inside the domination algorithm.

- In fiber two, $v_{2,1}$ is the only vertex not dominated; fiber three consists of $v_{2+1,1} = v_{3,1}$.
- In fiber three, $u_{3,0}$ is the only vertex not dominated; fiber four consists of $u_{3+1,0} = u_{4,0}$.
- In fiber four, $v_{4,2}$ is the only vertex not dominated; fiber five consists of $v_{4+1,2} = v_{5,2}$.
- In fiber five, $u_{5,1}$ is the only vertex not dominated; fiber six consists of $u_{5+1,1} = u_{6,1}$.
- In fiber six, $v_{6,0}$ is the only vertex not dominated. Although the next fiber would be equal to fiber one so now the loop stops.

It took six fibers to create a cycle so in the expression $(C_3 \times P_2 \times P_n)$, $n = 6$ for a minimal dominating set.

In Figure 4 we apply the domination algorithm (Algorithm 2) to the prism graph $(C_4 \times P_2)$. Starting with fiber one, we apply the clockwise algorithm (Algorithm 1) starting with $v_{1,0}$, then the second vertex will be $u_{1,3}$. Since $m = 4$, the clockwise algorithm will always produce a set with $\lceil \frac{4}{3} \rceil = 2$ vertex in each fiber. The next step in the domination algorithm is to construct fiber two using the clockwise algorithm starting at $v_{2,2}$ and goes to $u_{2,1}$. Because $m \equiv 1 \pmod{3}$, we need to remove the last vertex leaving fiber two as just $v_{2,2}$. Now we enter the while loop inside the domination algorithm.

- In fiber two, $u_{2,0}$ and $u_{2,1}$ are the not dominated vertices; fiber three consists of $u_{2+1,0} = u_{3,0}$ and $u_{2+1,1} = u_{3,1}$.
- In fiber three, $v_{3,3}$ is the only vertex not dominated; fiber four consists of $v_{3+1,3} = v_{4,3}$.
- In fiber four, $v_{4,1}$ and $u_{4,2}$ are the not dominated vertices; fiber five consists of $v_{4+1,1} = v_{5,1}$ and $u_{4+1,2} = u_{5,2}$.
- In fiber five, $u_{5,0}$ is the only vertex not dominated; fiber six consists of $u_{5+1,0} = u_{6,0}$.
- In fiber six, $v_{6,2}$ and $v_{6,3}$ are the not dominated vertices; fiber seven consists of $v_{6+1,2} = v_{7,2}$ and $v_{6+1,3} = v_{7,3}$.

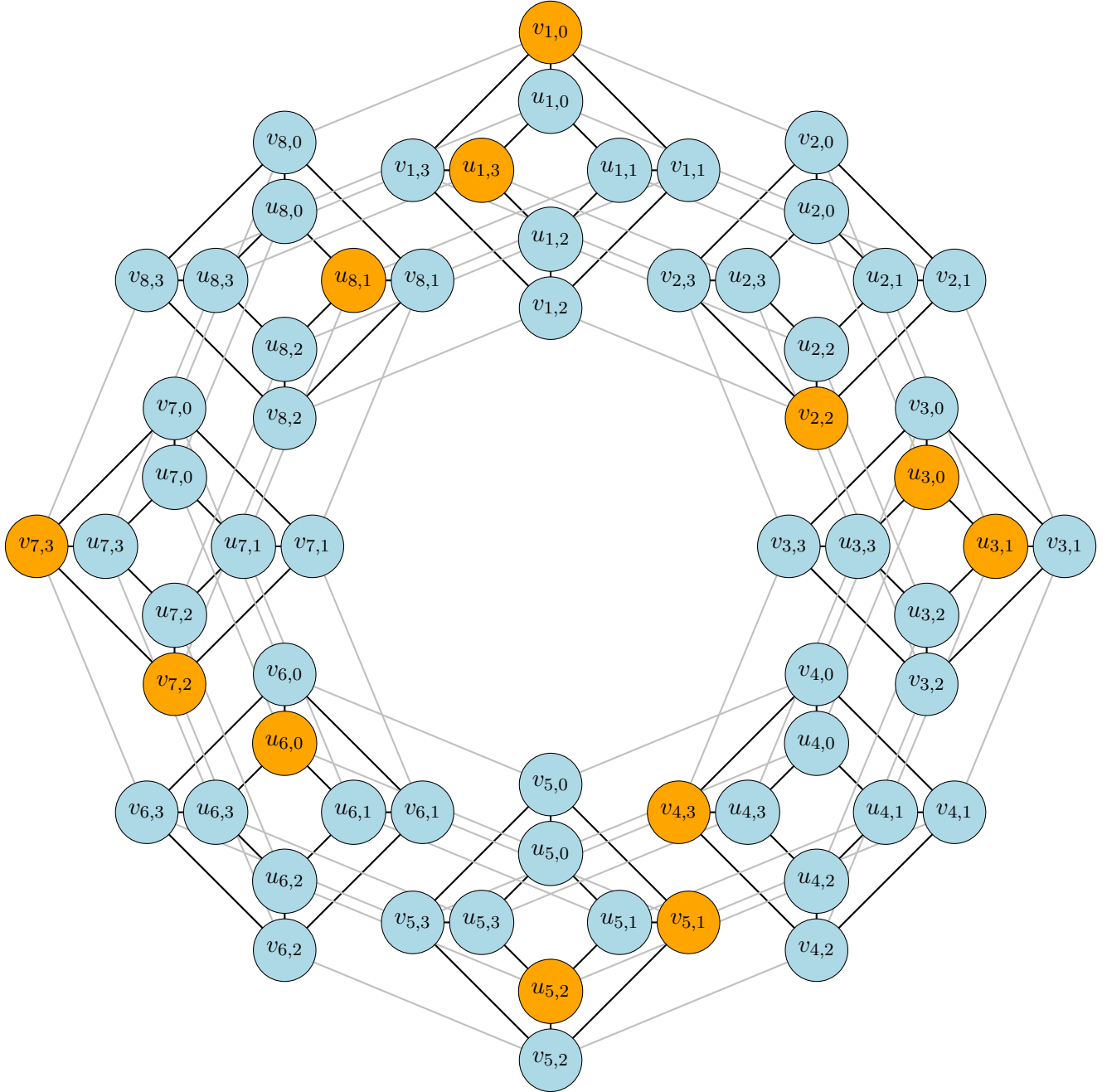


Figure 4: Application of Domination Algorithm (Algorithm 2 on $m = 4$)

- In fiber seven, $u_{7,1}$ is the only vertex not dominated; fiber eight consists of $u_{7+1,1} = u_{8,1}$.
- In fiber eight, $v_{8,0}$ and $u_{8,3}$ are the not dominated vertices. Although the next fiber would be equal to fiber one so now the loop stops.

It took eight fibers to create a cycle so in the expression $(C_4 \times P_2 \times P_n)$, $n = 8$ for a minimal dominating set.

In Figure 5 we apply the $0, 2, 5 \bmod 6$ domination algorithm (Algorithm 3) to the prism graph $(C_5 \times P_2)$.

- Fiber one, we apply the clockwise algorithm (Algorithm 1) starting with $v_{1,0}$. Since $m = 5$, the clockwise algorithm will always produce a set with $\lceil \frac{5}{3} \rceil = 2$ vertices in each fiber. By the clockwise algorithm, the second vertex of fiber one is $u_{1,(0+3) \bmod 5} = u_{1,3}$.
- Fiber two, we apply the clockwise algorithm starting with $v_{2,2}$. By the clockwise algorithm, the second vertex of fiber two is $u_{2,(2+3) \bmod 5} = u_{2,0}$.
- Fiber three, we apply the clockwise algorithm starting with $u_{3,1}$. By the clockwise algorithm, the second vertex of fiber three is $v_{3,(1+3) \bmod 5} = v_{3,4}$.

It took three fibers to create a cycle so in the expression $(C_5 \times P_2 \times P_n)$, $n = 3$ for a minimal dominating set.

Conjecture 2.3. *The dominating algorithms (Algorithm 2, Algorithm 3) always finds a minimum dominating set. The minimum dominating set repeats based on Table 3.*

[From Jonah] Dom025 has been tested up to $m = 1002$, and it works! We've also tested Dom0134 up to $m = 381$ and worked.

Conjecture 2.4. *The dominating set found by the dominating algorithms (Algorithm 2, Algorithm 3) is unique up to symmetry.*

[From Jonah] This is true based on 4, 5, and 8. Nothing else has been confirmed because they are BIG.

Lemma 2.4. *The $0, 2, 5$ domination algorithm (Algorithm 3) produces the "right" number of fibers.*

[From Jonah] The proof below still needs to be edited.

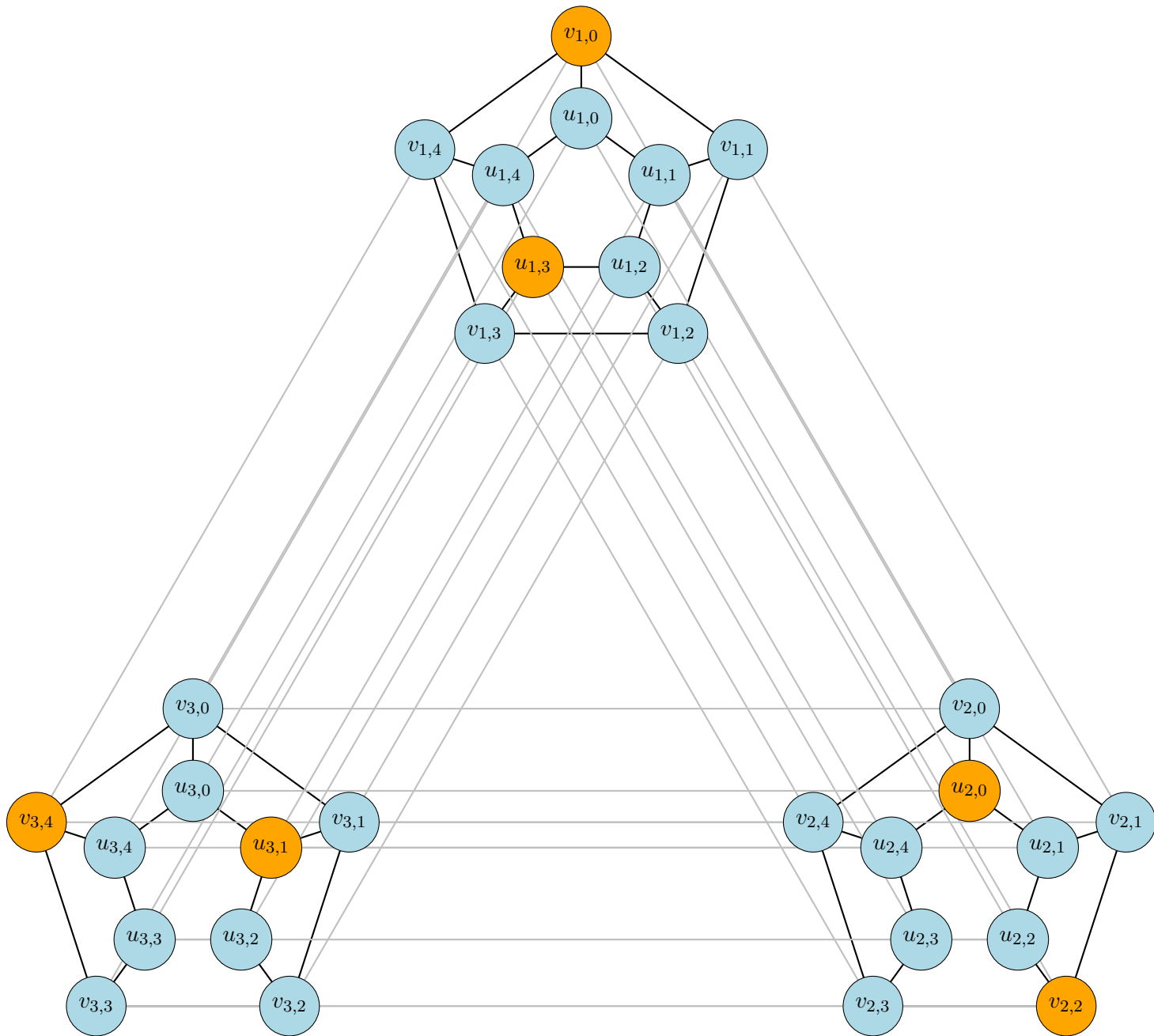


Figure 5: Application of Domination Algorithm (Algorithm 3) on $m = 5$

Proof. Let D be the set of vertices defined by Algorithm 1 and Algorithm 3 for $m \equiv 0 \pmod{6}$. Let $v_{r,s}$ be an arbitrary vertex on the exterior cycle. If $v_{r,s} \in D$, then $v_{r,s}$ is dominating. Suppose not, $v_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s+1}$

For all s , $v_{r,s}$ is dominated.

Let $u_{r,s}$ be an arbitrary vertex on inner cycle. If $u_{r,t} \in D$, then $u_{r,s}$ is dominated. Suppose not, $u_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t+1}$
- If $s \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$

For all t , $u_{r,t}$ is dominated. Since for every possible v and u as long as the clockwise algorithm is used as instructed by the 0, 2, 5 domination algorithm, the vertices in set D will be a dominating set. $\square$

Proof. Let D be the set of vertices defined by Algorithm 1 and Algorithm 3 for $m \equiv 2 \pmod{6}$. Let $v_{r,s}$ be an arbitrary vertex on the exterior cycle. If $v_{r,s} \in D$, then $v_{r,s}$ is dominating. Suppose not, $v_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$. (This is an exception since it continues the clockwise to $6k + 2 = m \equiv 0 \pmod{m}$.)
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s+1}$

For all s , $v_{r,s}$ is dominated.

Let $u_{r,s}$ be an arbitrary vertex on inner cycle. If $u_{r,t} \in D$, then $u_{r,s}$ is dominated. Suppose not, $u_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t+1}$
- If $s \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$

For all t , $u_{r,t}$ is dominated. Since for every possible v and u as long as the clockwise algorithm is used as instructed by the 0, 2, 5 domination algorithm, the vertices in set D will be a dominating set. $\square$

Proof. Let D be the set of vertices defined by Algorithm 1 and Algorithm 3 for $m \equiv 5 \pmod{6}$. Let $v_{r,s}$ be an arbitrary vertex on the exterior cycle. If $v_{r,s} \in D$, then $v_{r,s}$ is dominating. Suppose not, $v_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $s \equiv 0 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r-1,s}$
- If $s \equiv 1 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s+1}$
- If $s \equiv 2 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s}$.
- If $s \equiv 3 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r,s-1}$
- If $s \equiv 4 \pmod{6}$, $v_{r,s}$ is dominated by $v_{r+1,s}$
- If $s \equiv 5 \pmod{6}$, $v_{r,s}$ is dominated by $u_{r,s+1}$

For all s , $v_{r,s}$ is dominated.

Let $u_{r,s}$ be an arbitrary vertex on inner cycle. If $u_{r,t} \in D$, then $u_{r,s}$ is dominated. Suppose not, $u_{r,s}$ could be in one of three copies of $Y_{k,2}$.

Case 1: $r \equiv 0 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t+1}$
- If $s \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$

Case 2: $r \equiv 1 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$

Case 3: $r \equiv 2 \pmod{3}$.

- If $t \equiv 0 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t-1}$
- If $t \equiv 1 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r+1,t}$
- If $t \equiv 2 \pmod{6}$, $u_{r,t}$ is dominated by $v_{r,t}$
- If $t \equiv 3 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r-1,t}$
- If $t \equiv 4 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t+1}$
- If $t \equiv 5 \pmod{6}$, $u_{r,t}$ is dominated by $u_{r,t}$. (This is an exception since it continues the clockwise to $6k + 5 = m \equiv 0 \pmod{m}$.)

For all $t, u_{r,t}$ is dominated. Since for every possible v and u as long as the clockwise algorithm is used as instructed by the 0, 2, 5 domination algorithm, the vertices in set D will be a dominating set. $\square$

Lemma 2.5. *0134Domination produces the "right" numbers of vertices.*

Proof. This proof will take two cases based on the value of m .

- First Case: $m \equiv 0, 3 \pmod{6}$, thus $|fiber_k| = \frac{m}{3}$.

Using the clockwise algorithm (Algorithm 1) to construct $fiber_1$ and $fiber_2$, we have $|fiber_1| = |fiber_2| = \frac{m}{3}$. We need to show that $|fiber_3| = \frac{m}{3}$. By the 0, 1, 3, 4, domination algorithm (Algorithm 2), we construct $fiber_3$ from the undominated vertices in fiber two. When $m \equiv 0, 3 \pmod{6}$ the clockwise algorithm will space out the chosen vertices so that there is no overlap in vertices that are dominated and each vertex dominates itself and three other vertices. Since there are $\frac{m}{3}$ vertices chosen, this means that fiber two has $\frac{4m}{3}$ vertices that are currently dominated. By construction of $fiber_1$ and $fiber_2$, another $\frac{m}{3}$ dominated vertices from $fiber_1$ are added to the set of dominated vertices in fiber two, thus the total amount of dominated vertices in fiber two is $\frac{5m}{3}$. Since there are $2m = \frac{6m}{3}$ vertices in fiber two, that leaves $\frac{6m}{3} - \frac{5m}{3} = \frac{m}{3}$ vertices undominated in fiber two. Therefore, $|fiber_3| = \frac{m}{3}$ and those vertices follow the clockwise algorithm.

$Fiber_1$ and $fiber_2$ were carefully constructed using the clockwise algorithm and gave rise to $fiber_3$ which has $\frac{m}{3}$ vertices and follows the clockwise pattern. If we did the same process on $fiber_{k-1}$ and $fiber_k$ to create a $fiber_{k+1}$, it would also have $\frac{m}{3}$ vertices and follow the clockwise pattern. Inductively, all next fibers will have $\frac{m}{3}$. Thus when $m \equiv 0, 3 \pmod{6}$, the domination algorithm produces fibers with $|fiber_k| = \frac{m}{3}$.

- Second Case: $m \equiv 1, 4 \pmod{6}$, thus $|fiber_{2k}| = \frac{m-1}{3}$ and $|fiber_{2k+1}| = \frac{m+2}{3}$.

Using the clockwise algorithm to construct $fiber_1$ and $fiber_2$, we have $|fiber_1| = \frac{m+2}{3}$ and $|fiber_2| = \frac{m-1}{3}$. We need to show that $|fiber_3| = \frac{m+2}{3}$. By the 0, 1, 3, 4, domination algorithm, we construct $fiber_3$ from the undominated vertices in fiber two. When $m \equiv 1, 4 \pmod{6}$, the clockwise algorithm will space out the chosen vertices in even fibers so that there is no overlap in vertices that are dominated and each vertex dominates itself and three other vertices. Since there are $\frac{m-1}{3}$

vertices chosen, this means that fiber two has $\frac{4m-4}{3}$ vertices that are currently dominated. By construction of $fiber_1$ and $fiber_2$, another $\frac{m+2}{3}$ dominated vertices from $fiber_1$ are added to the set of dominated vertices in fiber two, thus the total amount of dominated vertices in fiber two is $\frac{5m-2}{3}$. Since there are $2m = \frac{6m}{3}$ vertices in fiber two, that leaves $\frac{6m}{3} - \frac{5m-2}{3} = \frac{m+2}{3}$ vertices undominated in fiber two. Therefore, $|fiber_3| = \frac{m+2}{3}$.

We need to show that $|fiber_4| = \frac{m-1}{3}$. By the 0, 1, 3, 4, domination algorithm, we construct $fiber_4$ from the undominated vertices in fiber three. There are $\frac{m+2}{3}$ vertices chosen in $fiber_3$, however the first and last vertices are adjacent meaning two fewer vertices are dominated. This means that fiber three has $4\frac{m+2}{3} - 2 = \frac{4m+2}{3}$ vertices that are currently dominated. By construction of $fiber_2$ and $fiber_3$, another $\frac{m-1}{3}$ vertices from $fiber_2$ are added to the set of dominated vertices in $fiber_3$, thus the total amount of dominated vertices in fiber three is $\frac{5m+1}{3}$. Since there are $2m = \frac{6m}{3}$ vertices in fiber three, that leaves $\frac{6m}{3} - \frac{5m+1}{3} = \frac{m-1}{3}$ vertices undominated in fiber three. Therefore, $|fiber_4| = \frac{m-1}{3}$.

$Fiber_1$ and $fiber_2$ were carefully constructed using the clockwise algorithm and gave rise to $fiber_3$ which has $\frac{m+2}{3}$ vertices. Similarly, $fiber_2$ and $fiber_3$ gave rise to $fiber_4$ which has $\frac{m-1}{3}$ vertices. If we did the same process on $fiber_{2k-1}$ and $fiber_{2k}$ to create a $fiber_{2k+1}$, it would also have $\frac{m+2}{3}$ vertices. If we did the same process on $fiber_{2k}$ and $fiber_{2k+1}$ to create a $fiber_{2k+2}$, it would also have $\frac{m-1}{3}$ vertices. Inductively, all future fibers will follow the pattern $|fiber_{2k}| = \frac{m-1}{3}$ and $|fiber_{2k+1}| = \frac{m+2}{3}$.

□

Lemma 2.6. *The 0, 1, 3, 4 domination algorithm produces a dominating set.*

Proof. Pick a vertex $w_{r,k}$ in fiber r . There are three cases to consider depending on the location of the vertex.

- Case 1: The vertex is in the closed neighborhood of fiber r : $w_{r,k} \in N[fiber_r]$.

If it is in the neighborhood of the set, then that is what it means to be dominated by the set.

- Case 2: The vertex is in the neighborhood of fiber $r - 1$: $w_{r,k} \in N[fiber_{r-1}]$

If it is in the neighborhood of the set, then that is what it means to be dominated by the set.

- Case 3: The vertex is not in the neighborhood of fiber $r - 1$ or fiber r : $w_{r,k} \notin N[fiber_{r-1}] \cup N[fiber_r]$

In this case the vertex is currently undominated. The algorithm says $w_{r+1,k}$ will be included in $fiber_{r+1}$. Thus, $w_{r,k} \in N[fiber_{r+1}]$, thus it is dominated.

Every vertex is dominated.

□

References

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